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- **Problem Statement ID – 25258**
- **Problem Statement title- Real-Time Monitoring System for Disaster Management Trainings**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 25258**
- **Team Name - Disaster x**

DISASTER
X

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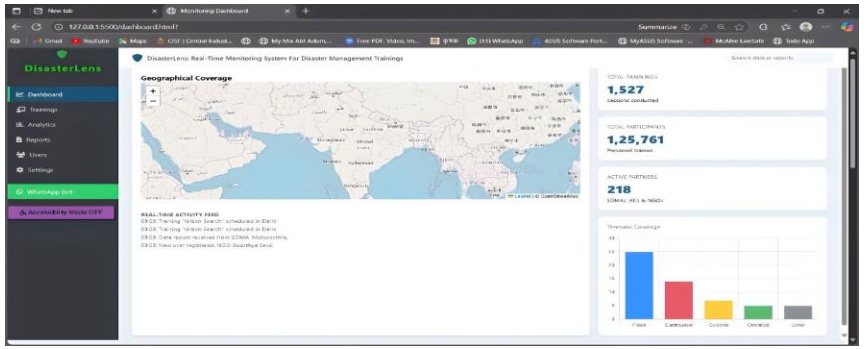
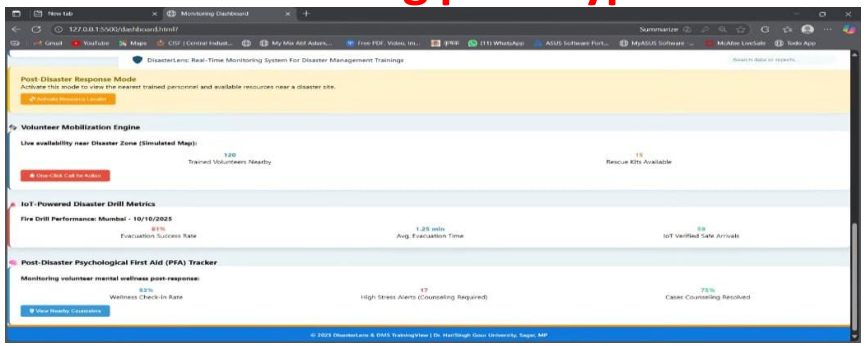
A centralized digital platform (web + mobile) named "**Disaster Lens**" for real-time monitoring, coordination, and evaluation of all disaster management trainings under NDMA . It integrates GIS visualization, AI analytics, face recognition, QR scanning, and What App automation to ensure seamless data flow and rapid communication across all stakeholders.

- **Centralized Dashboard:** Real-time tracking for NDMA, SDMAs, and training institutes.
- **Face Recognition & QR Attendance :** Smart, instant verification of participants.
- **What App BOT** : Sends alerts, updates, and feedback forms; enables one-click volunteer mobilization.
- **AI Analytics:** Detects skill gaps and predicts training needs.
- **Secure Access:** Role-based authentication ensures data privacy.
- **GIS Visualization:** Maps training locations, attendance, and progress.
- **Automated Reporting:** Generates instant performance summaries.
- **Mobile & IOT Integration:** Field data capture with geo-tagged attendance and report



Working prototype



DISASTER X

TECHNICAL APPROACH

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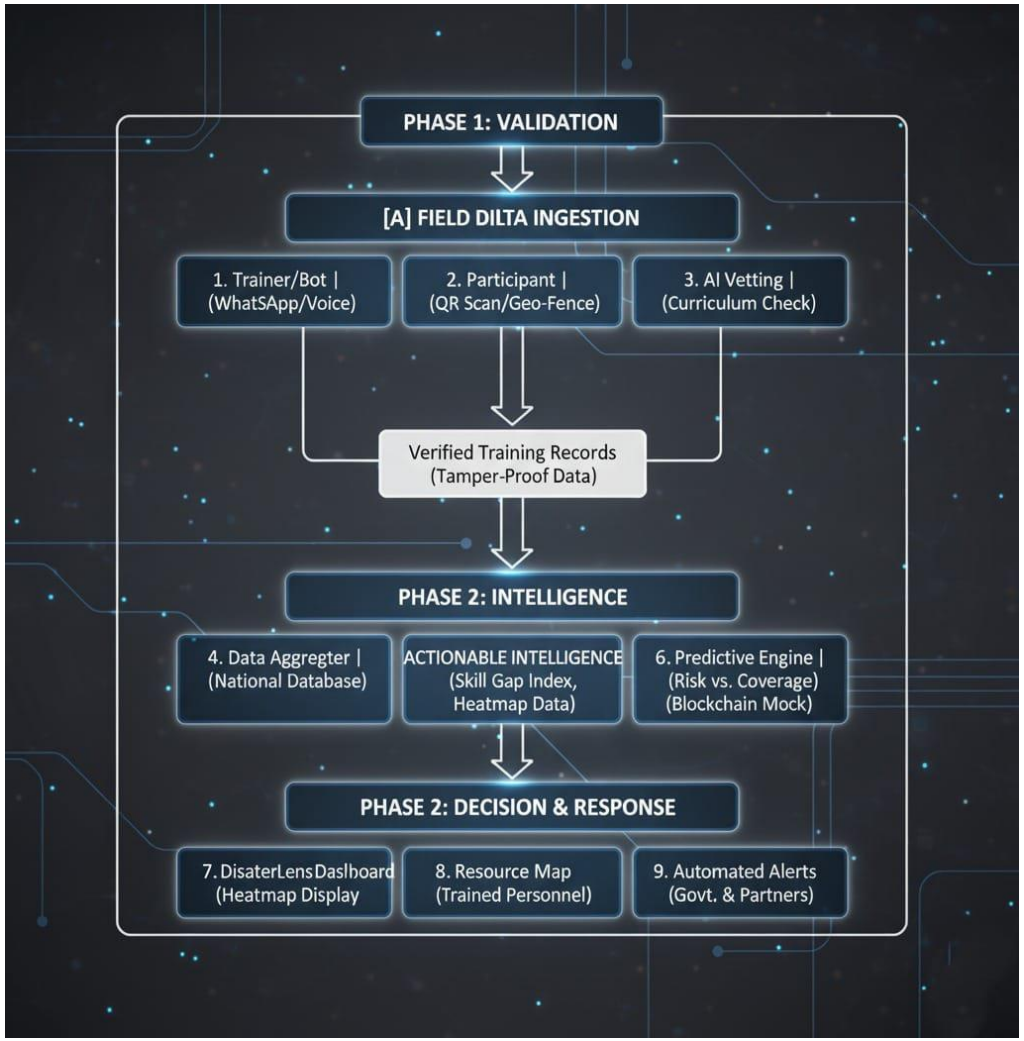

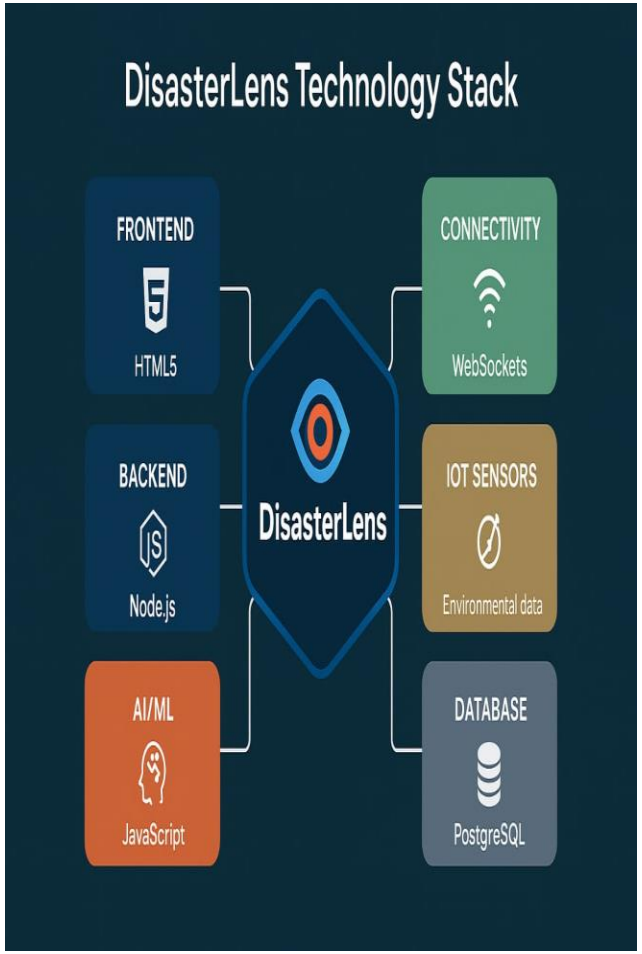

DISASTER X

FEASIBILITY AND VIABILITY

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⚖️ FEASIBILITY :

- **Technical feasibility**: Smart web-mobile System with real time data, cloud hosting, and secure access.
- **Operational feasibility**: Easy to use , work even offline, and smoothly fits existing NDMA workflows.
- **Economic Feasibility**: Built with open-source tech cost-effective, scalable , and sustainable.

📈 VIABILITY :

- **Social viability :** Boosts Disaster awareness and builds a culture of preparedness in communities.
- **Institutional Viability :** Easily adoptable by NDMA and state agencies with data-driven decision support.
- **Future Viability :** Scalable for AI analytics , real-time disaster response , and national expansion.

🏆 CHALLENGES :

- Lack of real-time visibility in training progress.
- Manual and delayed reporting from training center.
- Poor coordination between NDMA, SDMA and institutions.
- Limited internet in remote training areas.
- Difficulty in tracking training effectiveness.

💡 SOLUTION :

- Live dashboards with instant data updates and alerts.
- Automated digital reporting using cloud-based systems.
- Centralized platform enabling role-based access and collaboration.
- Offline data sync via progressive web app(PWA) support.
- AI-driven analytics for performance and impact assessment.

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IMPACT AND BENEFITS

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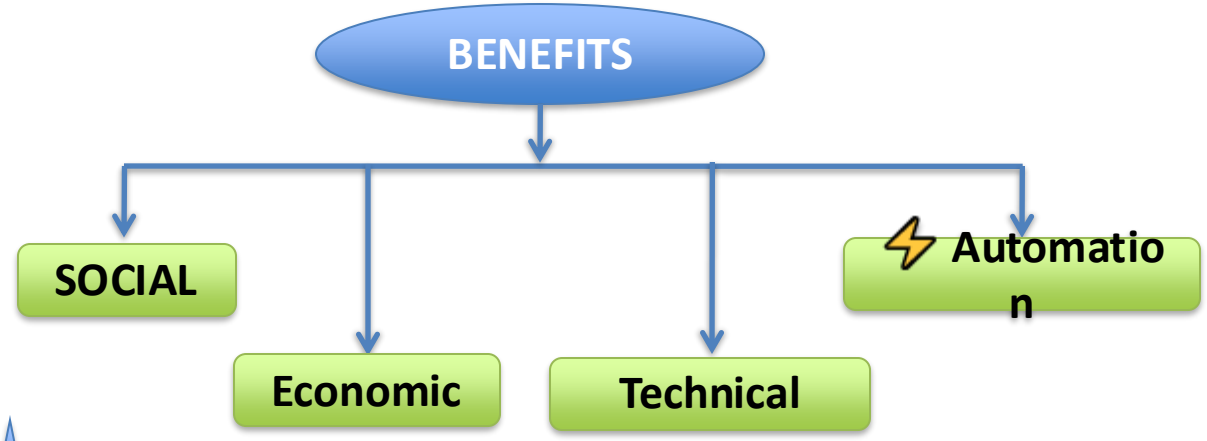

Potential impacts :

•**Government officials(NDMA,SDMA):** gain real time visibility of all training program for faster, data-driven decision.

•**Training institutes & instruction**: Simplified digital reporting, attendance tracking and performance evaluation.

•**Community volunteers & responders :** better access to training opportunities and continues skill improvement.

•**Local communities :** Increased disaster awareness, preparedness, and confidence during emergencies.

•**Policy makers & planner :** reliable data insights to identify training gaps and design effective strategies.

Benefits :

- **Social benefits:** Enhance public awareness and community preparedness promote citizen participation in disaster response training.
- **Economic benefits:**
- Built using open-source technologies – low implementation cost , long term cost efficient solution for govt. agencies.
- **Technical benefits:**

✓ Real time monitoring and automatic reporting.

✓ Centralized cloud based system ensure scalability

✓ Role based access control for secure data flow.

- **Strategic &long term benefits :**

✓ Crete a national training database for future disaster planning

✓ Enable AI-driven prediction and performance evaluation.

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RESEARCH AND REFERENCES

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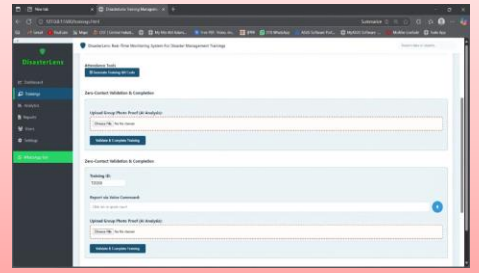
LINK OF OUR WORKING PROTOTYPE(LIVE HOSTED)

CLICK HERE

https://disasterlens.netlify.app/



Research and References links :

- National Disaster Management Authority (NDMA), India- https://ndma.gov.in
- National Institute of Disaster Management (NIDM) – https://nidm.gov.in
- Ministry of Home Affairs (MHA)- https://mha.gov.in
- United Nation Office for Disaster Risk Reduction (UNDRR)- https://www.undrr.org
- GIS and Data Visualization Tools-Arc GIS, Google Maps API, Power BI documentation (For integrating real time mapping and analytics)